Polarized closure of the integral Einstein obstruction

Gauge input, uniform constants, and all Einstein signs

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1 Statement and analytic dependency

Let X be a closed connected oriented smooth four-manifold, let

$$T_X = 2\chi(X) + 3\tau(X) > 0, \qquad M = X\#k\overline{\mathbb{CP}}^2\#\ell(S^1 \times S^3), \qquad D = k + 4\ell.$$

Assume that a Spin^c structure with real first Chern class c has a nonzero ordinary Seiberg–Witten invariant, with its insertion retained if the expected dimension is positive. When $b_+(X) = 1$, nonvanishing is in the small-perturbation chamber, not an arbitrarily chosen chamber. Write $A = c^2 \geq T_X > 0$, and define

$$\mathcal{E}(g) = \inf_{\lambda \neq 0} \frac{\int_M |\operatorname{Ric}_g - \lambda g|_g^2 \, d\mu_g}{\lambda^2 \operatorname{Vol}_g(M)}.$$

Stability theorem, relative to the integral calibration lemma below. There is a universal constant $\varepsilon > 0$ such that every smooth metric on M satisfies both

$$\mathcal{E}(g) + \left(\frac{T_X - D}{A} - \frac{2}{3}\right)_+ \geq \varepsilon, \qquad \mathcal{E}(g) + \left(\frac{1}{3} - \frac{D}{T_X}\right)_+ \geq \varepsilon. \tag{1}$$

In particular, if X is any minimal complex surface of general type, including $b_+(X) = 1$, then $A = T_X = c_1^2(X) > 0$, and

$$\boxed{\mathcal{E}(g) + \left(\frac{1}{3} - \frac{k + 4\ell}{c_1^2(X)}\right)_+ \geq \varepsilon.} \tag{2}$$

This note proves the gauge-to-obstruction implication, including its constants and the zero Einstein constant. Its nonclassical analytic input is stated precisely in Section 3. The full proof of that lemma remains the integral calibration manuscript; restating it here is not an independent verification of its Hessian and weak-Bianchi arguments. Neither a universal supply of special monopole classes on arbitrary four-manifolds nor positivity of a universal special-class invariant is assumed.

2 The actual polarized class and every auxiliary weight

2.1 Sign selection and the chamber

Fix g, and let $H = H_g^+$ be its positive harmonic subspace. Pull c back to M. Choose exceptional generators $E_{j,H}$, **with signs chosen at this** H, so that

$$E_{i,H} \cdot E_{j,H} = -\delta_{ij}, \qquad c \cdot E_{j,H} = 0, \qquad c_H^+ \cdot E_{j,H} \le 0.$$

Polarized Seiberg–Witten gluing supplies an actual Spin^c structure with

$$a_H = c - \sum_j E_{j,H} \tag{3}$$

and nonzero invariant at H. Thus its unperturbed equations are solvable for every H-adapted metric. The classical input is the construction in LeBrun's Proposition 3.2 [L01], not a consequence merely of the following intersection-form calculation:

$$\begin{aligned} (a_H^+)^2 &= (c_H^+)^2 - 2\sum_j c_H^+ \cdot E_{j,H} + \left|\sum_j E_{j,H}^+\right|^2 \ge (c_H^+)^2 \ge A, \\ a_H^2 &= A - k \ge T_X - k - 4\ell = T_M. \end{aligned} \tag{4}$$

For $b_+ = 1$, choose the positive generator ω_H so that $c \cdot \omega_H > 0$. This is possible because $c^2 > 0$. The sign selection implies

$$a_H \cdot \omega_H = c \cdot \omega_H - \sum_j E_{j,H} \cdot \omega_H > 0.$$

The corresponding base projection of ω_H has positive square. Hence the small-perturbation chamber agrees with the one in the chamber-compatible blow-up formula. Handle additions do not change the intersection form. If the base invariant uses an insertion z, retain it and add the ℓ degree-one loop insertions. Indeed,

$$\frac{a_H^2 - T_M}{4} = \frac{A - T_X}{4} + \ell.$$

The surgery and insertion statements used here are classical gauge theory [OS]; this dimension equality alone would not prove nonvanishing.

For a minimal general-type complex surface, the canonical Spin^c structure has $c = -K_X$, $K_X^2 > 0$, and $K_X \cdot [\omega] > 0$ for a Kähler form. When $b_+ = 1$, put K_X in the forward cone. Its pairing with every unit forward positive generator is at least $\sqrt{K_X^2} > 0$. The zero-perturbation discriminant for $-K_X$ is therefore negative, identifying the small chamber with the canonical symplectic chamber where nonvanishing holds [LL]. This is why $b_+ = 1$ is included. A symplectic structure without this chamber verification would not suffice.

2.2 Conformal covariance, with the bundle identification explicit

Let $\mathfrak{q}_g$ denote the quadratic spinor-to-self-dual-two-form map, normalized by $|\mathfrak{q}_g(\Phi)|^2 = |\Phi|^4/8$. For any smooth $f > 0$, set

$$\hat{g} = f^{-2}g = e^{2u}g, \qquad u = -\log f.$$

The Hodge star on two-forms is unchanged, so $H^+_{\hat{g}} = H^+_g = H$. Use the **same** Spin^c structure from (3) to solve the unperturbed equations for $\hat{g}$:

$$D^{\hat{g}}_{\mathcal{A}}\widehat{\Phi} = 0, \qquad -iF^+_{\mathcal{A}} = \mathfrak{q}_{\hat{g}}(\widehat{\Phi}).$$

Let $\beta : S_g \to S_{\hat{g}}$ be the fibrewise unitary spinor identification. The conformal identities are

$$D^{\hat{g}}_{\mathcal{A}}(e^{-3u/2}\beta\Phi) = e^{-5u/2}\beta D^g_{\mathcal{A}}\Phi, \qquad \mathfrak{q}_{\hat{g}}(\beta v) = e^{2u}\mathfrak{q}_g(v),$$

where the second equality compares covariant two-forms. With $\Phi = f^{-3/2}\beta^{-1}\widehat{\Phi}$, they give

$$D^g_{\mathcal{A}}\Phi = 0, \qquad \mathfrak{q}_g(\Phi) = f^{-1}\mathfrak{q}_{\hat{g}}(\widehat{\Phi}), \qquad -iF^+_{\mathcal{A}} = f\mathfrak{q}_g(\Phi). \tag{5}$$

This is the conformal computation underlying [L06, Lemma 3.7], applied only to the one conformal class needed here. The exact order is

$$\forall g\ \exists \mathfrak{s}_g\ \forall f \in C^\infty(M, (0,\infty))\ \exists(\mathcal{A}_f, \Phi_f) \text{ solving (5)}. \tag{6}$$

No continuity of $g \mapsto \mathfrak{s}_g$ is required. The class may change between metrics; it is held fixed while f varies. Ordinary solvability just at g would not establish (6).

3 The analytic input and the uniform constant

All Ricci norms below are tensor Hilbert–Schmidt norms. The norm on W^+ is its Hilbert–Schmidt norm as an endomorphism of Λ^+. Put $V = \mathrm{Vol}(g)$, $T = T_M$, and

$$\mathscr{O} = \mathrm{Orb}\,\mathrm{diag}(-1/6, 1/12, 1/12), F = \mathrm{dist}(W^+, \mathscr{O})^2, \qquad q = -s + \sqrt{6}|W^+|, \qquad p = (a^+_H)^2,$$
$$\Delta = \int q_+^2 - 72\pi^2 p, \qquad \kappa = \frac{1}{V}\left(\int (|\,\mathrm{Ric} + g|^2 + F) + \Delta\right), \qquad B = \frac{(-a^2_H)_+}{V}.$$

Integral calibration lemma (the nonclassical input). For an actual structure satisfying (6) and $p > 0$, the signed spectral monopole inequality gives $\Delta \geq 0$, and a universal $c_{\mathrm{cal}} > 0$ satisfies

$$(1 + B)\kappa \geq c_{\mathrm{cal}}. \tag{7}$$

The proof in the integral calibration manuscript constructs bounded weighted spinors, obtains a covariant Hessian estimate using integral Ricci and connection curvature, forms a smooth polynomial Weyl tensor, and tests weak Bianchi against its divergence. Its hypotheses involve the selected structure at the metric under consideration, not an all-metric invariant. This note uses (7) explicitly; it does not replace that proof by an appeal to a previous round's conclusion.

We next give the complete constant transfer. Put

$$r = V^{-1} \int |\operatorname{Ric} + g|^2, \qquad d = (T/A - 2/3)_+.$$

The algebraic comparison detailed in the appendix yields

$$\kappa \le \frac{245}{2} r + 972\pi^2 \frac{A}{V} d. \tag{8}$$

The scalar monopole bound gives

$$32\pi^2 A \le \int s_-^2 \le \int s^2 \le 32V + 8rV, \qquad A/V \le (4 + r)/(4\pi^2).$$

Consequently $r \le 1$ implies $\kappa \le 1215(r + d)$. For $r \le 1/4$, Chern–Weil and the reverse triangle inequality give

$$\begin{aligned} 4\pi^2 T/V &= V^{-1} \int (s^2/24 + 2|W^+|^2 - |\operatorname{Ric}^\circ|^2/2) \\ &\ge (4 - 2\sqrt{r})^2/24 - r/2 \ge 1/4. \end{aligned} \tag{9}$$

Since $a_H^2 \ge T > 0$, we have $B = 0$ in this branch, even when the family allows arbitrarily many blow-ups. Equations (7)–(9) imply

$$r + d \ge \varepsilon, \qquad \varepsilon = \min\{1/4, c_{\mathrm{cal}}/1215\} > 0. \tag{10}$$

For $r > 1/4$, (10) is immediate. This proves uniformity with respect to the manifold, the metric, the number of exceptional classes, and their selection. Smooth approximations of the auxiliary weight are taken with g, a_H fixed before their approximation error tends to zero; no uniform rate of smoothing across metrics is used.

For the positive normalization, set $r_+ = V^{-1} \int |\operatorname{Ric} - g|^2$. On the scalar-negative set, $s_-^2 \le (s-4)^2$, whence

$$A/V \le r_+/(8\pi^2).$$

If $r_+ \le 1/4$ and $d \le 1/4$, the analogue of (9) gives $4\pi^2 T/V \ge 1/4$, whereas $T \le A(2/3 + d)$ gives

$$4\pi^2 T/V \le (r_+/2)(2/3 + d) \le 11/96 < 1/4.$$

Thus $r_+ + d > 1/4$. For any $\lambda \neq 0$, apply the appropriate normalization to $\tilde{g} = |\lambda| g$. Constant rescaling preserves H, (4), and the weighted input. In dimension four,

$$\frac{1}{V_{\tilde{g}}} \int |\operatorname{Ric}_{\tilde{g}} - \operatorname{sign}(\lambda)\tilde{g}|^2_{\tilde{g}} = \frac{\int |\operatorname{Ric}_g - \lambda g|^2_g}{\lambda^2 V_g}.$$

Taking the infimum proves the first inequality in (1). Apply the same argument with the smaller positive projection budget $T_X \leq A$ to obtain the second. No comparison of ratios with a potentially negative numerator is necessary. Equation (2) follows by $A = T_X$.

4 Scalar-flat exclusion and all Einstein constants

With $f = 1$, the spinor Weitzenböck formula, integrated on the closed manifold, gives

$$0 = \int (4|\nabla_{\mathcal{A}}\Phi|^2 + s|\Phi|^2 + |\Phi|^4).$$

Writing $\rho = |\Phi|^2$, Cauchy–Schwarz and the curvature equation yield

$$\int s_-^2 \geq \int \rho^2 = 8 \int |F^+_{\mathcal{A}}|^2 \geq 32\pi^2 (a^+_H)^2 \geq 32\pi^2 A > 0. \tag{11}$$

The spinor cannot vanish identically because $p > 0$. In particular, there are no scalar-flat metrics in this family, for any k, ℓ.

For $\operatorname{Ric}_g = \lambda_0 g$ with $\lambda_0 \neq 0$, the definition gives $\mathcal{E}(g) = 0$. For a Ricci-flat metric it instead gives $\mathcal{E}(g) = 4$, since $|\lambda g|^2_g = 4\lambda^2$. The Ricci-flat case is therefore excluded by (11), not by assigning it zero defect.

Choose $\gamma = \varepsilon/2$. From (1) and (11),

$$\boxed{D \geq (1/3 - \gamma) T_X \quad \Longrightarrow \quad M \text{ admits no Einstein metric of any sign.}} \tag{12}$$

The sharper necessary inequality for a nonzero Einstein constant is $D/T_X \leq 1/3 - \varepsilon$. The closed threshold (12) uses a smaller constant intentionally; equality at $1/3 - \varepsilon$ is not excluded by (1) alone.

For general-type X and $\ell = 0$, (12) contains every case of LeBrun's Corollary 3.4. In its closed band, every metric also satisfies

$$\mathcal{E}(g) \geq \varepsilon/2, \qquad \delta_\sigma(g) \geq \sqrt{\varepsilon/8}, \quad \delta_\sigma(g) = \|g^{-1} \operatorname{Ric}_g - \sigma I\|_{\infty,\mathrm{op}}, \quad \sigma = \pm 1,$$

because $\mathcal{E}(g) \leq 4\delta_\sigma(g)^2$.

5 Theorem 3.3: the positive budget and the zero corner

The positive-budget connected-sum setting of Theorem 3.3 is covered by (1) and (12), including its handle term and the specified chamber. No new gauge theorem is required beyond its polarized input.

For nonpositive budgets, use Chern–Weil directly. If $T_M < 0$, the identity in (9) excludes Einstein metrics. There is also an elementary integral bound that does not require any monopole input. Set

$$R_2 = \int |\operatorname{Ric}|^2, \quad S = \int s, \quad L = \int (s - S/V)^2, \quad W_2 = \int |W^+|^2.$$

When $R_2 > 0$, optimization in $1/\lambda$ gives

$$\mathcal{E} = 4 - \frac{S^2}{VR_2} = 1 + \frac{12W_2 - 24\pi^2 T_M + L}{R_2}. \tag{13}$$

Thus $T_M \leq 0$ implies $\mathcal{E} \geq 1$, with the case $R_2 = 0$ handled by $\mathcal{E} = 4$. This is a classical algebraic consequence, not a new calibration theorem.

If $T_X < 0$, or $T_X = 0 < D$, then $T_M = T_X - D < 0$, and the exact nonexistence conclusion follows. However, $T_X = D = 0$ **is different**: an Einstein metric is then forced to be Ricci-flat and anti-self-dual, not impossible. K3 with its complex orientation is an actual example. It has nonzero Seiberg–Witten invariant and admits Ricci-flat metrics.

The inspected author-hosted PDF and arXiv v1 of [L01] are the March 2000 preprint, not the journal typesetting. Their unrestricted wording of Theorem 3.3 must not be imported across this zero corner. The journal's publication metadata alone does not establish how that corner was worded in its final text. Our precise claim is the quantitative extension for $T_X > 0$, together with the classical $T_M < 0$ exclusion. This qualification does not narrow Corollary 3.4: there $T_X = c_1^2(X) > 0$.

6 What is strengthened

The additional threshold interval contains actual members of the general-type family. Let

$$X_m = \Sigma_2 \times \Sigma_m, \qquad A_m = 8(m-1), \qquad k_m = \lceil A_m/3 \rceil - 1, \qquad m \geq 2.$$

These are minimal general-type bases and

$$0 < 1/3 - k_m/A_m \leq 1/A_m \longrightarrow 0.$$

For all sufficiently large m, their blow-ups satisfy (12), although $k_m < A_m/3$. Their Euler characteristics tend to infinity. Thus the strengthening is not merely an enlargement of unattained real parameter values. This does not assert novelty over every other obstruction in the literature, nor produce an explicit first m without an explicit calibration constant.

The correct implication is **integral stability plus scalar-flat exclusion implies the classical exact obstruction**. The exact obstruction alone would not imply a uniform integral gap. LeBrun's

absolute Ricci-energy infima and minimal-volume results are different conclusions, not consequences established by this note.

7 Appendix: algebra behind the constant transfer

Write $e = s + 4$, $\alpha = \sqrt{2/3}|W^+|$, $w = \lambda_{\min}(W^+)$, $t = s + 24\alpha$, and

$$b = 2s/3 + 2w, \quad G = (2s/3 - 2\alpha)^2 - b_-^2 \geq 0, \quad P = \int |\operatorname{Ric}^\circ|^2.$$

The signed spectral monopole bound is $\int b_-^2 \geq 32\pi^2 p$. Here $v_+ = \max(v, 0)$ and $v_- = \max(-v, 0)$ for a real number v. The pointwise identity

$$t^2 + 18G = 216(s^2/24 + 2|W^+|^2) - 18b_-^2$$

therefore gives

$$\int (t^2 + 18G) \leq 108P + 864\pi^2 Ad.$$

Also $F \leq e^2 + t^2 + G$. One direct verification sets $z = w + \alpha \in [0, \alpha]$, so

$$F = \tfrac{3}{2}(\alpha - 1/6)^2 + z/2, \qquad \alpha = (t - e + 4)/24.$$

For $s \leq -3$, $b < 0$ and $G \geq 8z$, giving $F \leq (t^2 + e^2)/192 + G/16$. For $s > -3$, $e > 1$, and $z \leq \alpha$, $4 - e \leq 3e^2$, $t \leq (t^2 + 1)/2$ give $F \leq (t^2 + 5e^2)/64$. Hence

$$\int (|\operatorname{Ric} + g|^2 + F) \leq 109rV + 864\pi^2 Ad.$$

Finally, $27(s^2/24 + 2|W^+|^2) - q^2 = t^2/8$, so $q_+^2 \leq q^2$ and $p \geq A$ yield

$$\Delta \leq (27/2)rV + 108\pi^2 Ad.$$

Adding proves (8). These estimates involve only the selected class's projection bound and its square; neither a finite class set nor a continuous selection is hidden in the constants.

8 References and proof locations

- [L01] C. LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Invent. Math. **145** (2001), 279–316. Author-hosted preprint, Definition 1, Proposition 3.2, Theorem 3.3, Corollary 3.4; publication record.
- [L06] C. LeBrun, *Four-manifolds, curvature bounds, and convex geometry*, preprint, Lemma 3.7.
- [LL] T.-J. Li and A.-K. Liu, *Uniqueness of symplectic canonical class, surface cone and symplectic cone of 4-manifolds with* $b^+ = 1$, arXiv:math/0012048, Section 2.1.

- [OS] P. Ozsváth and Z. Szabó, *Higher type adjunction inequalities in Seiberg–Witten theory*, arXiv:math/0005268, Proposition 2.2, Remark 2.3, Theorem 3.2.

The package's analytic proof of (7) is Integral calibration, Sections 1–6; the exact spinor and weighted Hodge identities are in Monopole input, Sections 6–8. The earlier polarized proof and chamber analysis remain intact. The present note consolidates their interfaces; it is not a new independent main theorem or an audit of every analytic identity in those manuscripts.